CPTG: Redshift Stress Test of ZF-UDS-7329

Carter L Glass Jr

Abstract

ZF-UDS-7329, also identified as PRIMER-EXCELS-55410, is a massive compact quiescent galaxy observed at spectroscopic redshift $z = 3.1943 \pm 0.0003$. This article presents a self-contained Curvature Polarization Transport Gravity (CPTG) high-redshift stress test of ZF-UDS-7329 using its measured compact radius and stellar velocity dispersion. Using the EXCELS effective radius $R_c = 0.910 \pm 0.010\,\mathrm{kpc}$ and stellar dispersion $\sigma_\star = 250 \pm 20\,\mathrm{km\,s^{-1}}$ gives a local stellar support scale $a_\sigma = (2.226 \pm 0.357) \times 10^{-9}\,\mathrm{m\,s^{-2}}$. On the adopted flat redshift-stress reference branch, $E(z) = 4.7787$, $H(z) = 334.51\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and the observed epoch corresponds to $t_{\rm em} = 1.969\,\mathrm{Gyr}$ with lookback time $t_{\rm lookback} = 11.498\,\mathrm{Gyr}$. The corresponding expansion-normalized support scale is $a_{\sigma,z} = (4.6578 \pm 0.7470) \times 10^{-10}\,\mathrm{m\,s^{-2}}$. For the aperture-matched effective-radius branch, the CPTG field prediction differs from the observed dynamical comparator by $-1.10 \times 10^{-9}\,\mathrm{m\,s^{-2}}$, corresponding to an error-weighted offset of -0.43σ. ZF-UDS-7329 therefore provides a compact high-redshift CPTG stress test at an early cosmic epoch while remaining consistent with the observed effective-radius dynamical field.

1 Motivation

The CPTG [4, 5] redshift stress test asks whether the compact stellar support scale of a high-redshift galaxy is tested after normalization by the expansion factor at the epoch of observation. ZF-UDS-7329 is suitable for this calculation because the EXCELS/PRIMER analysis reports a secure spectroscopic redshift, a compact effective radius, a measured stellar velocity dispersion, and an effective-radius dynamical mass for the same object [1]. The organization follows the following stress-test form: define the local support scale from $\sigma_\star$ and R_c, place the galaxy at its observed cosmic epoch through $E(z)$ and $H(z)$, construct the time-linked acceleration branch, then compare the effective-radius CPTG output to the observed dynamical comparator with observational error.

2 Observed Inputs

The observational anchor is the EXCELS/PRIMER analysis of ZF-UDS-7329, listed as PRIMER-EXCELS-55410 in the EXCELS catalogue [1]. The quantities used directly in this article are the spectroscopic redshift, compact effective radius, integrated stellar velocity dispersion, stellar mass within the effective radius, and effective dynamical mass within the same aperture.

Quantity	Adopted value	Role in test
Formal target	ZF-UDS-7329 / PRIMER-EXCELS-55410	Object name
Redshift	$z = 3.1943 \pm 0.0003$	Expansion epoch
Stellar dispersion	$\sigma_\star = 250 \pm 20\,\mathrm{km\,s^{-1}}$	Support scale
Compact-radius interval	$R_c = 0.900$–$0.920\,\mathrm{kpc}$	Observed radius range
Adopted midpoint radius	$R_c = 0.910\,\mathrm{kpc}$	Reference calculation
Stellar mass within R_e	$\log_{10}(M_{\star,e}/M_\odot) = 10.84 \pm 0.03$	Aperture-matched source branch
Effective dynamical mass	$\log_{10}(M_{\mathrm{dyn,e}}/M_\odot) = 10.94 \pm 0.07$	Observed dynamical comparator

Table 1: Observed quantities used for the ZF-UDS-7329 redshift stress test and effective-radius CPTG comparison.

3 Parameter List and Aperture Convention

The calculation uses a compact set of parameters. The parameter list is included to keep the radius and acceleration symbols distinct without changing the notation used in the response equation.

Symbol	Meaning in this article	Value or use
z	Spectroscopic redshift of ZF-UDS-7329	3.1943 ± 0.0003
R_c	Compact observed radius used to define the stellar support scale	$0.910 \pm 0.010\,\mathrm{kpc}$
R_e	Effective-radius aperture used for the stellar-mass and dynamical-mass comparison	Same aperture scale as the compact radius in this calculation
$\sigma_\star$	Observed stellar velocity dispersion	$250 \pm 20\,\mathrm{km\,s^{-1}}$
$a_\sigma(R_c)$	Local stellar support scale from the measured compact radius and dispersion	$\sigma_\star^2/R_c$
$a_{\sigma,z}(R_c)$	Expansion-normalized redshift-stress comparator	$a_\sigma(R_c)/E(z)$
$a_\star$	Local structural acceleration scale used in the effective-radius CPTG response	Set equal to $a_\sigma(R_c)$ for the final response
$g_{\star,e}$	Aperture-matched stellar source field at the effective radius	$GM_{\star,e}/R_e^2$
$g_{\mathrm{dyn,e}}$	Observed effective-radius dynamical comparator	From $M_{\mathrm{dyn,e}}$ at the same aperture
R_C	CPTG geometric-remnant scale appearing only inside g_{geom}	Not the observed compact radius R_c

Table 2: Parameter list and aperture convention used in the ZF-UDS-7329 redshift stress test.

4 Local Stellar Support Scale

The compact support scale is defined by

$$a_\sigma(R_c) = \frac{\sigma_\star^2}{R_c}. \tag{1}$$

Using

$$\sigma_\star = 250 \pm 20\,\mathrm{km\,s^{-1}}, \qquad R_c = 0.910 \pm 0.010\,\mathrm{kpc}, \tag{2}$$

gives

$$a_\sigma = (2.2258 \pm 0.3570) \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{3}$$

At fixed midpoint radius, the dispersion brackets are

$$a_\sigma(230\,\mathrm{km\,s^{-1}}) = 1.8839 \times 10^{-9}\,\mathrm{m\,s^{-2}}, \tag{4}$$

and

$$a_\sigma(270\,\mathrm{km\,s^{-1}}) = 2.5962 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{5}$$

5 Expansion Factor and Gyr Comparison

For this ZF-UDS-7329 redshift-stress calculation, the reference branch is

$$E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}, \tag{6}$$

with

$$\Omega_m = 0.3, \qquad \Omega_\Lambda = 0.7, \qquad H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{7}$$

At the observed redshift,

$$z = 3.1943, \tag{8}$$

this gives

$$E(z) = 4.7787, \qquad H(z) = 334.51\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{9}$$

The corresponding flat-branch cosmic time is

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \sinh^{-1}\left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}(1+z)^{-3/2}\right]. \tag{10}$$

For ZF-UDS-7329,

$$t_{\mathrm{em}} = 1.969\,\mathrm{Gyr}, \qquad t_0 = 13.467\,\mathrm{Gyr}, \qquad t_{\mathrm{lookback}} = 11.498\,\mathrm{Gyr}. \tag{11}$$

Epoch	Redshift	Time (Gyr)	Lookback (Gyr)
Observed epoch	3.1943	1.969	11.498
Present epoch	0	13.467	0.000

Table 3: Gyr comparison for the adopted flat redshift-stress reference branch.

6 Time-Linked CPTG Acceleration Scale

The expansion-normalized support scale is

$$a_{\sigma,z}(R_c) = \frac{a_\sigma(R_c)}{E(z)}. \tag{12}$$

For the adopted midpoint radius,

$$a_{\sigma,z} = (4.6578 \pm 0.7470) \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{13}$$

The time-linked branch is written as

$$a_\star(t; R_c) = a_{\star,0}(R_c)E[z(t)], \tag{14}$$

with

$$a_{\star,0}(R_c) = a_{\sigma,z}(R_c). \tag{15}$$

At the observed epoch,

$$a_\star(t_{\mathrm{em}}; R_c) = a_\sigma(R_c) = 2.2258 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{16}$$

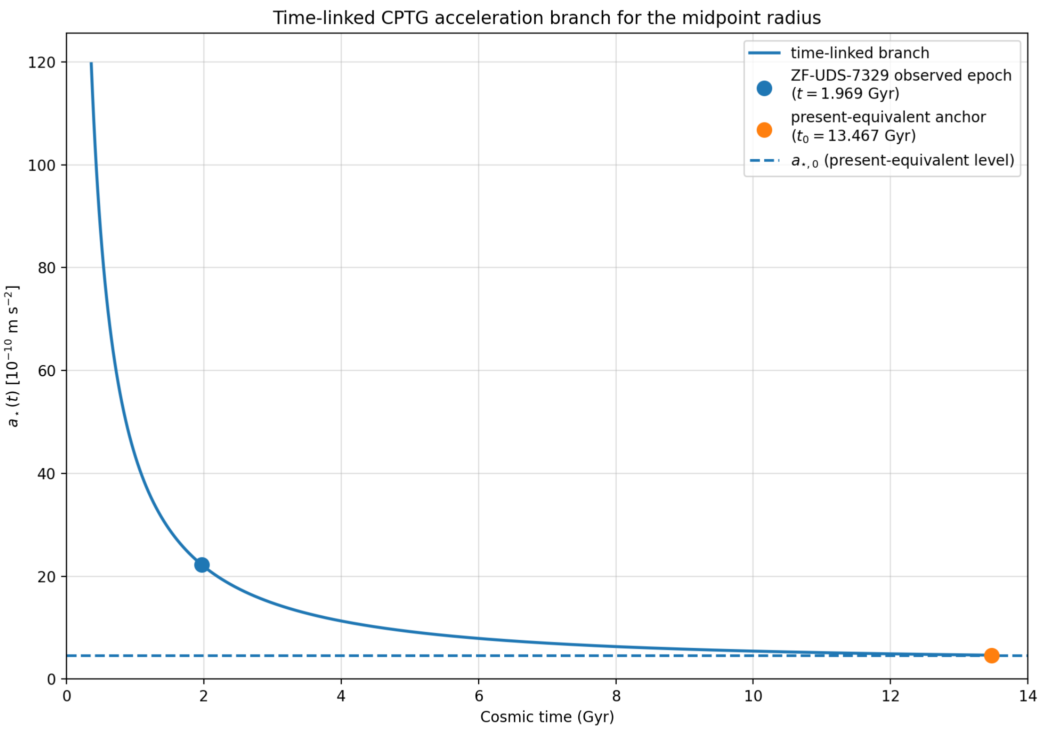

Figure 1: Time-linked CPTG acceleration branch for the midpoint radius. The curve follows the redshift-stress organization $a_\star(t; R_c) = a_{\star,0}(R_c)E[z(t)]$. The blue point marks the observed epoch of ZF-UDS-7329, and the orange point marks the present-equivalent anchor for the same midpoint radius.

7 Radius-Scan Redshift Stress Scale

For the midpoint dispersion, the radius-scan form is

$$a_{\sigma,z}(R_c) = \frac{\sigma_\star^2}{R_c E(z)}. \tag{17}$$

Across the adopted compact-radius interval,

$$a_{\sigma,z}(0.900\,\mathrm{kpc}) = 4.7095 \times 10^{-10}\,\mathrm{m\,s^{-2}}, \tag{18}$$

$$a_{\sigma,z}(0.910\,\mathrm{kpc}) = 4.6578 \times 10^{-10}\,\mathrm{m\,s^{-2}}, \tag{19}$$

and

$$a_{\sigma,z}(0.920\,\mathrm{kpc}) = 4.6071 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{20}$$

At fixed midpoint radius, the dispersion bracket gives

$$a_{\sigma,z}(230\,\mathrm{km\,s^{-1}}) = 3.9423 \times 10^{-10}\,\mathrm{m\,s^{-2}}, \tag{21}$$

and

$$a_{\sigma,z}(270\,\mathrm{km\,s^{-1}}) = 5.4328 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{22}$$

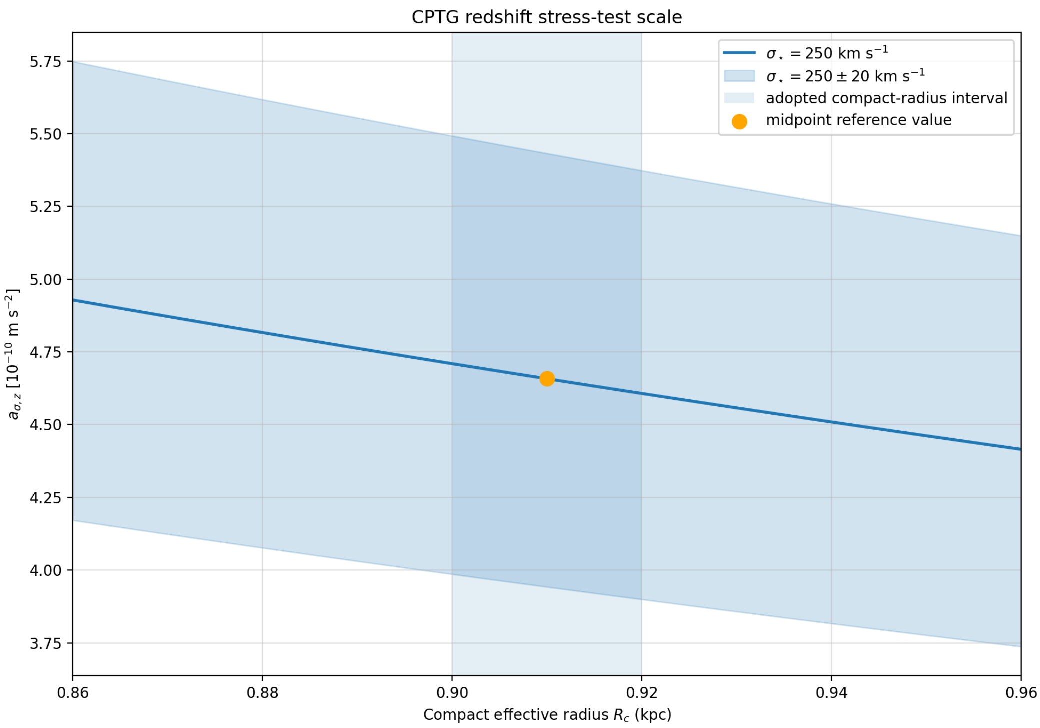

Figure 2: CPTG redshift stress-test scale as a function of compact effective radius. The curve shows the expansion-normalized support scale inferred from the measured stellar dispersion, while the surrounding band propagates the published dispersion uncertainty. The shaded vertical region marks the adopted compact-radius interval, and the orange point gives the midpoint value used for the article's reference calculation.

8 Final Comparison to Observed Dynamics

The preceding sections define the redshift-normalized stress scale. The final step is separate: it checks whether the same object's aperture-matched stellar source and effective-radius dynamical comparator are consistent with the reduced CPTG response at R_e.

For this final response, the local structural acceleration scale is

$$a_\star \equiv a_\sigma(R_c) = 2.2258 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{23}$$

The expansion-normalized value $a_{\sigma,z}$ is the redshift-stress comparator and time-branch anchor; it is not the acceleration scale inserted into the effective-radius response equation. The aperture-matched stellar source field entering the response is

$$g_{\star,e}(R_e) = \frac{GM_{\star,e}}{R_e^2} = (1.165 \pm 0.084) \times 10^{-8}\,\mathrm{m\,s^{-2}}. \tag{24}$$

The reduced CPTG response is solved from

$$g = g_\star + \frac{\sqrt{a_\star g_\star} + g_{\mathrm{geom}}}{\left[1 + \left(\frac{g}{a_\star}\right)^{123/50}\right]^{11/50}}, \tag{25}$$

with

$$g_{\mathrm{geom}}(r) = \eta a_\star \frac{(r/R_C)^{7/5}}{1 + (r/R_C)^{12/5}}, \qquad \eta = \frac{1}{2}, \qquad R_C = \frac{48\pi c^2}{a_\star}. \tag{26}$$

Using the aperture-matched source field, local structural scale, and midpoint radius gives the CPTG effective-radius field

$$g_{\mathrm{CPTG,e}}(R_e) = (1.356 \pm 0.091) \times 10^{-8}\,\mathrm{m\,s^{-2}}. \tag{27}$$

The observed dynamical comparator is

$$g_{\mathrm{dyn,e}} = (1.466 \pm 0.238) \times 10^{-8}\,\mathrm{m\,s^{-2}}. \tag{28}$$

Therefore the field difference is

$$g_{\mathrm{CPTG,e}}(R_e) - g_{\mathrm{dyn,e}} = -1.10 \times 10^{-9}\,\mathrm{m\,s^{-2}}, \tag{29}$$

with error-weighted offset

$$\Delta_{g,e} = -0.43\sigma. \tag{30}$$

In mass space,

$$\log_{10}(M_{\mathrm{CPTG,e}}/M_\odot) = 10.906 \pm 0.028, \quad \log_{10}(M_{\mathrm{dyn,e}}/M_\odot) = 10.940 \pm 0.070, \tag{31}$$

so that

$$\log_{10}(M_{\mathrm{CPTG,e}}/M_\odot) - \log_{10}(M_{\mathrm{dyn,e}}/M_\odot) = -0.034\,\mathrm{dex}, \tag{32}$$

with

$$\Delta_{M,e} = -0.45\sigma. \tag{33}$$

Quantity	Difference	Error-weighted offset
$g_{\mathrm{CPTG,e}}(R_e) - g_{\mathrm{dyn,e}}$	$-1.10 \times 10^{-9}\,\mathrm{m\,s^{-2}}$	-0.43σ
$\log_{10}(M_{\mathrm{CPTG,e}}/M_\odot) - \log_{10}(M_{\mathrm{dyn,e}}/M_\odot)$	$-0.034\,\mathrm{dex}$	-0.45σ

Table 4: Final comparison to the observed effective-radius dynamical quantities.

9 Conclusion

ZF-UDS-7329 provides a compact high-redshift CPTG stress test because the same object has a secure spectroscopic redshift, a measured compact radius, a measured integrated stellar velocity dispersion, and an effective-radius dynamical comparator. The local support scale from the observed compact radius and dispersion is

$$a_\sigma = (2.226 \pm 0.357) \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{34}$$

On the adopted flat redshift-stress reference branch, the galaxy is observed at $E(z) = 4.7787$ and $H(z) = 334.51\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, corresponding to $t_{\mathrm{em}} = 1.969\,\mathrm{Gyr}$ and $t_{\mathrm{lookback}} = 11.498\,\mathrm{Gyr}$. Dividing the local support scale by the expansion factor gives the present-equivalent stress scale

$$a_{\sigma,z} = (4.6578 \pm 0.7470) \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{35}$$

The effective-radius comparison then uses the local structural scale $a_\star = a_\sigma(R_c)$ together with the aperture-matched stellar source field $g_{\star,e} = GM_{\star,e}/R_e^2$. This separates the redshift-stress comparator from the final response calculation while keeping both tied to the same observed compact system.

The resulting CPTG effective-radius field is

$$g_{\mathrm{CPTG,e}}(R_e) = (1.356 \pm 0.091) \times 10^{-8}\,\mathrm{m\,s^{-2}}, \tag{36}$$

compared with the observed dynamical field

$$g_{\mathrm{dyn,e}} = (1.466 \pm 0.238) \times 10^{-8}\,\mathrm{m\,s^{-2}}. \tag{37}$$

The direct field difference is

$$g_{\mathrm{CPTG,e}}(R_e) - g_{\mathrm{dyn,e}} = -1.10 \times 10^{-9}\,\mathrm{m\,s^{-2}}, \tag{38}$$

with error-weighted offset $\Delta_{g,e} = -0.43\sigma$. In mass space, the offset is

$$\log_{10}(M_{\mathrm{CPTG,e}}/M_\odot) - \log_{10}(M_{\mathrm{dyn,e}}/M_\odot) = -0.034\,\mathrm{dex}, \tag{39}$$

with $\Delta_{M,e} = -0.45\sigma$. Within the scope of this article, ZF-UDS-7329 therefore supplies both a stable redshift-normalized compact support scale and an effective-radius CPTG response that remains within the published observational uncertainty of the dynamical comparator.

References

[1] Carnall, A. C., et al. The JWST EXCELS survey: too much, too young, too fast? Ultra-massive quiescent galaxies at $3 < z < 5$. Monthly Notices of the Royal Astronomical Society 534, 325–348 (2024). DOI: 10.1093/mnras/stae2092. ADS: 2024MNRAS.534..325C. arXiv: 2405.02242. Publisher: Oxford Academic.

[2] Turner, C., et al. Age-dating early quiescent galaxies: high star formation efficiency, but consistent with direct, higher-redshift observations. Monthly Notices of the Royal Astronomical Society 537, 1826–1848 (2025). DOI: 10.1093/mnras/staf128. ADS: 2025MNRAS.537.1826T. arXiv: 2410.05377. Publisher: Oxford Academic.

[3] Glazebrook, K., et al. A massive galaxy that formed its stars at $z \approx 11$. Nature 628, 277–281 (2024). DOI: 10.1038/s41586-024-07191-9. ADS: 2024Natur.628..277G. arXiv: 2308.05606. Publisher: Nature.

[4] Carter L Glass Jr, Curvature Polarization Transport Gravity: A Unified Geometric Framework for Cosmic Structure and Expansion, DOI: 10.13140/RG.2.2.26030.68164.

[5] CPTG, Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity, companion resource for the present work, available at `https://github.com/CLG2025/CPTG`.